HACKATHON PROJECT PRESENTATION

AI-Powered Personal Health Copilot

Your Smart Digital Health Companion



OCR + AI | 5 languages | Voice AI



TEAM: NEXGEN SOLUTIONS

Muthayammal Engineering College

02 / PROBLEM STATEMENT

The Problem

Medical information is scattered across prescriptions, blood reports, diagnostic reports and discharge summaries.



Medical jargon
Patients struggle to understand medical terminology.

Hidden abnormal values
Out-of-range test values are easy to miss.

No organized history
Records stay scattered across many documents.

Proposed Solution

An AI-powered healthcare web application that converts medical documents into structured, easy-to-understand health information.

Read with OCR

Reads PDF, DOC, DOCX, JPG, JPEG and PNG medical documents.

Extract with AI

Finds medicines, dosages, test values, dates and diagnoses.

Explain simply

Explains report findings in simple language.

Organize history

Builds a unified health profile from every record.

Document Intelligence

AI Medical OCR

Reads PDF, DOC, DOCX, JPG, JPEG and PNG documents.

Medical Data Extraction

Medicines, dosages, test values, units, ranges, dates and diagnoses.

AI Health Summary

Explains medical documents in simple language.

Abnormal Value Detection

Highlights values outside the reference ranges in reports.

Current Health Summary

Latest available medical information, up to the current date.

Unified Health Profile

Documents, medications, results and history in one place.

Health Timeline and Calendar

Medical history in order, with upload dates tracked.

Health Analytics and Reminders

Visualize available health trends and manage follow-ups.

05 / KEY FEATURES (2 OF 2)

Access and Experience

AI Health Copilot

Answers questions about authorized records and guides navigation.

Multilingual Website

English, Tamil, Hindi, Telugu and Malayalam across the interface.

Thanglish Support

Natural Tamil-English conversations with the AI assistant.

Voice AI

Voice input and spoken replies where speech services support the language.

Patient and Admin Portals

Role-based dashboards, patient management and secure record access.

Sign Up and Sign In

Patients register, get a unique Patient ID and see only their records.

Bulk Document Upload

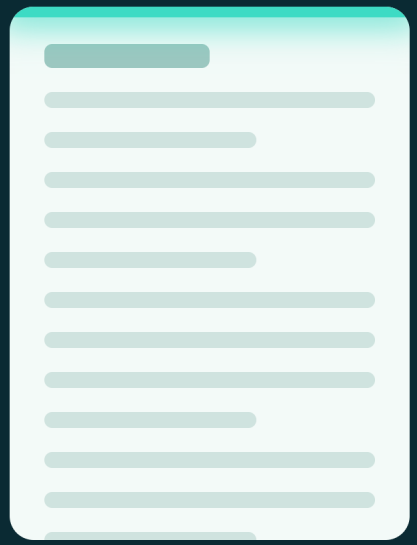
Admins upload many documents and map them to the right patients.

ABHA/FHIR Readiness

FHIR-inspired data model and mock ABHA connectivity.

How It Works

1. Upload a medical document
2. Validate the file and extract text with OCR
3. AI identifies relevant medical information
4. Validate and store data in a structured database
5. Generate a simple-language summary
6. Highlight abnormal values using reference ranges
7. Update profile, timeline and current health summary
8. AI Copilot answers from authorized information only



System Architecture

STEP 1
Medical Document Upload

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STEP 2
File Validation

→

STEP 3
OCR

STEP 4
AI Medical Extraction

→

STEP 5
Data Validation

→

STEP 6
Database

STEP 7
AI Summary and Abnormal Detection

→

STEP 8
Patient Dashboard

→

STEP 9
Timeline, Calendar, Analytics, and AI Copilot

Technology Stack

FRONTEND
React, TypeScript, Vite, Tailwind CSS

BACKEND
Node.js, Express.js, TypeScript

DATABASE
MongoDB

AI
Large Language Model API

OCR
Document-processing service chosen by availability and accuracy

AUTHENTICATION
Secure, role-based authentication

INTERNATIONALIZATION
Centralized translation system

VOICE
Speech-to-text and text-to-speech services

DEPLOYMENT
Vercel, Render/Railway and MongoDB Atlas

Innovation and Uniqueness

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capabilities combined in one platform

- Medical document intelligence
- Multilingual healthcare explanations
- Thanglish conversations
- Voice interaction
- Current health summaries
- Patient-specific record management
- Bulk uploads
- FHIR-inspired data organization

Security and Medical Safety

Secure access

- Role-based access control
- Backend-enforced patient data isolation
- Patients cannot access other patients' records

Medical safety

- The AI does not diagnose diseases or prescribe treatments
- It explains information found in uploaded documents
- Users are advised to consult qualified healthcare professionals
- Synthetic patient data is used for demonstrations



Expected Impact

Less confusion
Reduce confusion around medical documents.

Clearer information
Improve access to understandable health information.

Simpler records
Simplify medical record management.

Organized journey
Help patients keep a more organized view of their healthcare journey.

More accessible, easier-to-understand healthcare information.

12 / FUTURE SCOPE

Future Scope

01
Authorized integration with ABDM/ABHA services

02
Broader FHIR interoperability

03
Improved OCR for handwritten and bilingual prescriptions

04
Additional Indian languages

05
Integration with authorized healthcare providers and wearable health data

Thank You

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AI-Powered Personal Health Copilot: your smart digital health companion.

